Candidate Boolean F4/F5 allocation audit

UTC run id: `run-20261011T1715Z`. Native run and comparison: completed, exit status 0.

All 72 requested candidate records completed and matched baseline outputs and non-allocation counters exactly. Native comparison admitted the frozen allocation thresholds. Observed allocation request reductions: F4 0.25%, F5 7.70%, criterion 35.12%. Cumulative requested-byte reductions: F4 42.47%, F5 45.47%, criterion 50.30%. Percentages are native derived values, 100 x (1 - candidate/baseline), over the 24 calls per arm. Every workload met or improved both allocation measures.

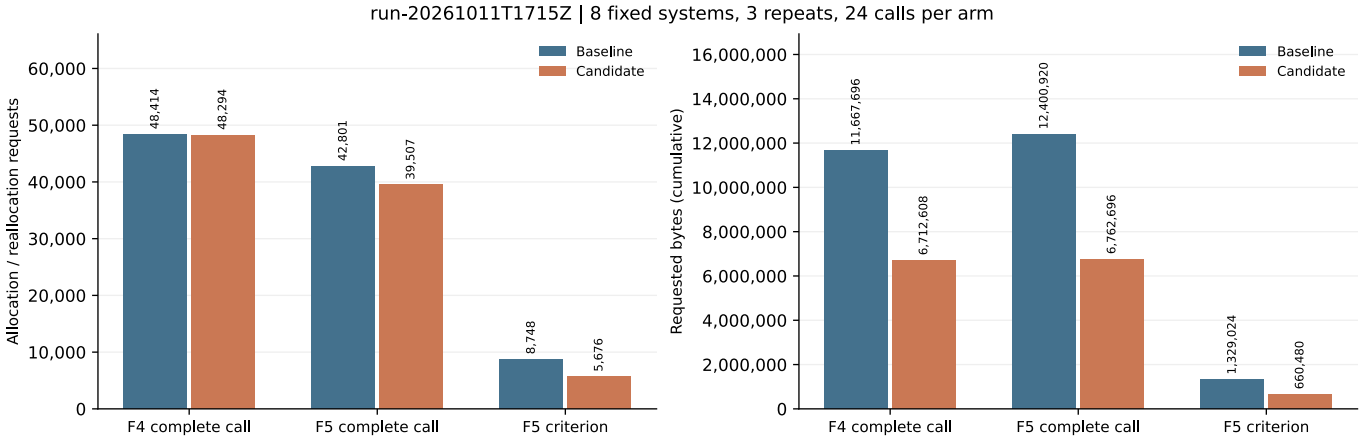


Figure 1: run-20261011T1715Z | 8 fixed systems, 3 repeats, 24 calls per arm. Recorded counts and cumulative requested bytes; 24 calls per arm.

Arm	Baseline requests	Other requests	Baseline bytes	Other bytes
criterion	8748	5676	1329024	660480
f4	48414	48294	11667696	6712608
f5	42801	39507	12400920	6762696

Candidate kernel revision: 2e3dffda681a4771be511653180da8354de3bce3. Baseline production: e957e48c2b8cf1f84a7fc44a1d673103964a789d, with the same formatted diagnostic. Source/binary hashes: records/candidate-sha256.txt. Raw records: records/candidate.jsonl; matched comparison: records/comparison.json; per-system values: records/per-system-counts.csv.

The first library-test build failed on two pre-existing fixtures: an undefined parser fixture and a removed historical sorting utility. Both were repaired in test-only code while keeping all assertions. Failed and successful logs: `records/f4-f5-candidate-test-build*.log`. All 73 focused F4/F5 and Macaulay tests passed before this audit. No candidate audit records failed.

Host and scope. Linux x86-64, AMD EPYC 9V74, 5 logical CPUs, 18,882,699,264 bytes RAM; Rust 1.99.0 (b940084d7), release, one Rayon worker. Exclusive CPU isolation was not established. Wall time and peak memory were not measured. This is a standalone Boolean matrix allocation diagnostic; full ECDLP cost, matched rho ratios, m=83 scaling and general-field gains remain unestablished.

Inputs and accounting. Eight planted quadratic systems: (n,m,d) = (8,8,3), (8,8,4), (10,10,4), (10,10,5), seeds `0x2545f4914f6cdd1d` and `0x9e3779b97f4a7c15`. Warmup precedes three repetitions. Each bar covers 24 calls. Fixture generation, hashing and verification are outside the audit; packing, elimination and output construction are included. Bytes are cumulative requests, not peak memory. Raw repeated values are retained; no statistical interval was estimated.

Evidence. Repository `aburan28/crypto`; study `research/f4_f5_row_build_20261011`. Native driver: `examples/f4_f5_row_build_bench.rs`. Full commands and paths: this directory's `REPORT.md`; graph values: `graph-data.json`; rendering source: `../../render_reports.py`; raw records and hashes: `../../records/`.